

Seaweed consumption and the risk of thyroid cancer in women: the Japan Public Health Center-based Prospective Study.

Iodine is a suspected risk factor for thyroid cancer. Seaweed accounts for about 80% of Japanese people's iodine intake. We examined the association between seaweed consumption and the risk of thyroid cancer in Japanese women. Women participating in the Japan Public Health Center-based Prospective Study (n=52 679; age: 40-69 years) were followed up for a mean of 14.5 years; 134 new thyroid cancer cases, including 113 papillary carcinoma cases, were identified. Seaweed consumption was assessed using a food-frequency questionnaire and divided into three categories: 2 days/week or less (reference); 3-4 days/week; and almost daily. The Cox proportional hazards model was applied to estimate hazard ratios (HRs) and 95% confidence intervals (CIs). Seaweed consumption was clearly associated with an increased risk of papillary carcinoma (HR for almost daily consumption compared with 2 days/week or less=1.71; 95% CI: 1.01-2.90; trend P=0.04). After stratification for menopausal status, an increased risk was observed in postmenopausal women (papillary carcinoma HR for almost daily consumption compared with 2 days/week or less=3.81, 95% CI: 1.67-8.68; trend P<0.01), but not in premenopausal women (HR=0.91, 95% CI: 0.44-1.91; trend P=0.76). This study identified a positive association between seaweed consumption and the risk of thyroid cancer (especially for papillary carcinoma) in postmenopausal women.